

Dustin Poston

LetsDefend

Rule: SOC113 - Suspicious hh.exe Usage

EventID : 44
Event Time : Jan, 31, 2021, 04:59 PM
Rule : SOC113 - Suspicious hh.exe Usage
Level : Security Analyst
Source Address : 172.16.17.47
Source Hostname : BillPRD
File Name : WinRAR.chm
File Hash : 07694464c25bac4ecdb365e928ffe1ff
File Size : 306 KB
Device Action : Allowed
File (Password:infected) : Download

High- Level Summary

This alert was generated for suspicious hh.exe usage involving a compiled HTML Help file on the endpoint BillPRD. The event details reviewed include EventID: 44, alert type: Malware, source IP: 172.16.17.47, source host: BillPRD, file name: WinRAR.chm, file hash: 07694464c25bac4ecdb365e928ffe1ff, file size: 306 KB, and event time: January 31, 2021 at 4:59 PM. Required further investigation to validate and confirm file is legitimate or needing immediate attention.

Verdict

Outcome concluded the alert was a False Positive.

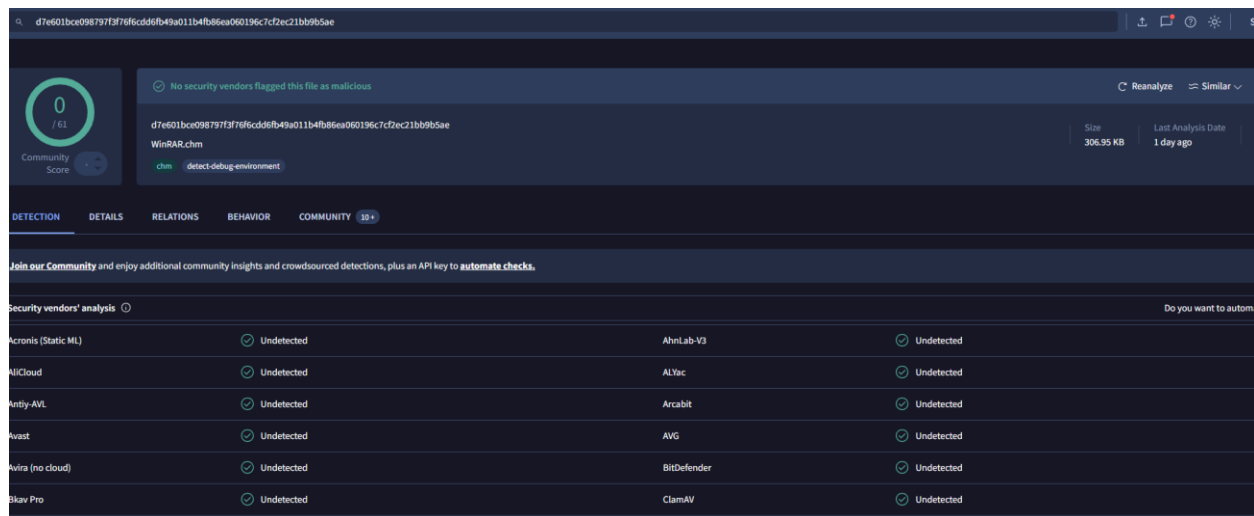
Timeline of Activity

The suspicious hh.exe usage was observed on January 31, 2021 at 4:59 PM. The activity involved the file WinRAR.chm executing or being opened through the HTML Help process on the host BillPRD. No additional suspicious activity was identified during the reviewed timeframe, and no outbound connections, quarantine events, or further endpoint concerns were observed.

DATE	TYPE	SRC ADDRESS	SRC PORT	DEST. ADDRESS	DEST. PORT
Oct, 11, 2020, 09:41 PM	Proxy	172.16.17.47	41221	172.217.17.238	443
Sep, 20, 2020, 10:54 PM	Firewall	172.16.17.47	54211	5.188.0.251	443
Sep, 20, 2020, 10:54 PM	Proxy	172.16.17.47	54211	5.188.0.251	443
Oct, 11, 2020, 09:39 PM	Proxy	172.16.17.47	23312	172.217.17.238	443
Oct, 11, 2020, 09:39 PM	Firewall	172.16.17.47	23312	172.217.17.238	443
Oct, 29, 2020, 07:05 PM	Proxy	172.16.17.47	39485	217.8.117.77	443
Oct, 29, 2020, 07:34 PM	Proxy	172.16.17.47	46938	115.99.150.132	56841
Jan, 02, 2021, 04:33 AM	Proxy	172.16.17.47	45827	67.199.248.10	443

Investigation of the Alert and Findings

I reviewed the alert details, including the event identifier, alert type, source IP, source host, file name, file hash, file size, and timestamp. The activity showed hh.exe interacting with the file WinRAR.chm on BillIPRD. Since CHM files are opened through hh.exe and can sometimes be abused by attackers, I treated the alert as suspicious until the supporting evidence could be reviewed. I first checked the file hash to determine whether the file had any known malicious reputation, and the hash did not return any flags as shown below.



The screenshot shows the VirusShare interface for a file analysis. The file is identified as 'WinRAR.chm' with a size of 306.95 KB and a last analysis date of 1 day ago. The file hash is 'd7e601bce098797f3f76f6cdd6fb49a011b4fb86ea060196c7cf2ec21bb9b5ae'. The interface indicates that no security vendors flagged this file as malicious. A table below shows the results of security vendors' analysis, all of which are 'Undetected'.

Security vendors' analysis			
Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
AlCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Avast	Undetected	AVG	Undetected
Avira (no cloud)	Undetected	BitDefender	Undetected
Bkav Pro	Undetected	ClamAV	Undetected

The hash given was then compared to legitimate hash given for the WinRAR.chm file and was identical.

Basic properties ⓘ	
MD5	07694464c25bac4ecdb365e928ffe1ff
SHA-1	ebe42dd5830aea54a21639cee011d0b67b93433e
SHA-256	d7e601bce098797f3f76f6cdd6fb49a011b4fb86ea060196c7cf2ec21bb9b5ae

After checking the hash, I reviewed the endpoint activity to identify whether the file or process attempted to communicate externally. No outbound connections were identified from the endpoint in relation to this alert. I also reviewed the endpoint history and did not observe suspicious follow-on behavior, additional malicious process activity, or any signs that the file caused further compromise. The action was allowed by the security tool, and there was no quarantine activity associated with the event.

Endpoint Information

Host Information

Hostname: BillPRD

Domain: LetsDefend

IP Address: 172.16.17.47

Bit Level: 64

OS: Windows 10

Primary User: Bill01

Client/Server: Client

Last Login: Jan, 31, 2021, 07:56 PM

Action

Containment: ☐

Processes 4

Network Action 3

Terminal History 1

Browser History 4

Results: 10

EVENT TIME	PROCESS ID	PROCESS NAME	PARENT PROCESS	COMMAND LINE
No Event Time	No Process ID	Chrome.exe	—	c:/program files (x86)/google/chr...
No Event Time	No Process ID	hh.exe	—	C:/Windows/hh.exe
20.09.2020 14:51	No Process ID	AcroRd32.exe	—	c:/program files (x86)/adobe/acro...
20.09.2020 22:51	No Process ID	Krankheitsmeldung_092020_07.x...	—	c:/users/Bill/desktop/Krankheits...

Hypothesis: Suspicious CHM File Execution

During the investigation, I mapped the suspicious hh.exe usage to the MITRE ATT&CK framework under T1218.001, Compiled HTML File, because adversaries may abuse CHM files and the legitimate hh.exe process to conceal or launch malicious content.

Home > Techniques > Enterprise > System Binary Proxy Execution > Compiled HTML File

System Binary Proxy Execution: Compiled HTML File

Other sub-techniques of System Binary Proxy Execution (14)

Adversaries may abuse Compiled HTML files (.chm) to conceal malicious code. CHM files are commonly distributed as part of the Microsoft HTML Help system. CHM files are compressed compilations of various content such as HTML documents, images, and scripting/web related programming languages such as VBA, JScript, Java, and ActiveX. [1] CHM content is displayed using underlying components of the Internet Explorer browser [2] loaded by the HTML Help executable program (hh.exe). [3]

A custom CHM file containing embedded payloads could be delivered to a victim then triggered by User Execution. CHM execution may also bypass application application control on older and/or unpatched systems that do not account for execution of binaries through hh.exe. [4] [5]

ID: T1218.001

Sub-technique of: T1218

Tactic: Stealth

Platforms: Windows

Contributors: Rahmat Nurfauzi, @infosec1nja, PT Xynexis International

Version: 3.0

Created: 23 January 2020

Last Modified: 12 May 2026

Version

Permalink

In this case, the activity involved a CHM file named WinRAR.chm, which aligned with the type of behavior that can trigger this detection. However, after reviewing the hash reputation, network activity, endpoint history, and quarantine status, there was not enough evidence to support malicious execution. The most accurate hypothesis was that the alert was triggered by benign compiled help-file usage rather than malware activity.

Indicators of Compromise (IOC)

The key indicators reviewed during this investigation were source IP 172.16.17.47, source host BillPRD, file name WinRAR.chm, file hash 07694464c25bac4ecdb365e928ffe1ff, file size 306 KB, and EventID 44. No outbound connections were observed, the file hash produced no malicious flags, endpoint history appeared clear, and there was no quarantine activity.

Case Management Summary and Conclusion

I concluded in case management that the alert did not represent confirmed malware activity. The investigation showed that the file hash was clean, no outbound connections were made, endpoint history did not show suspicious behavior, and no quarantine occurred. Based on these findings, the alert should be documented as reviewed and closed as a false positive with no additional remediation required.

Recommendations

Recommendations include using execution prevention where applicable to reduce the chance of hh.exe being misused. If hh.exe is not required on a given system or network, application control should be considered to prevent it from executing and limit the opportunity for adversaries to abuse it. I would also recommend restricting web-based content by blocking the download, transfer, and execution of uncommon file types that are often seen in adversary campaigns, such as CHM files. These mitigations would help lower the risk of similar alerts becoming malicious in the future, while still allowing the current case to be closed as a false positive based on the investigation findings.